

Solutions to Problem Set 2

1. The birthday problem

Consider $n \in \mathbb{N}$ people assumed to have the same probability of being born on any day of the calendar year. Assume also for simplicity that this calendar consists of 365 days. We are interested in the event

$$A_n = \text{"at least 2 people in the group have the same birthday"}.$$

The aim of this exercise is to find bounds for $\mathbf{P}[A_n^c]$ for any fixed $2 \leq n \leq 365 = N$.

- (a) Write down an appropriate probability space $(\Omega, \mathcal{F}, \mathbf{P})$ for this "random experiment". Represent A_n^c as a subset of Ω .
- (b) Show that for any two given people i and j ($1 \leq i \neq j \leq n$) from the group,

$$\mathbf{P}[i \text{ and } j \text{ have the same birthday}] = \frac{1}{N}.$$

- (c) Using (b), and the known inequality

$$\log(1+x) \leq x, \quad x \in (-1, \infty)$$

show that

$$1 - \frac{n(n-1)}{2N} \leq \mathbf{P}[A_n^c] \leq \exp\left(-\frac{n(n-1)}{2N}\right).$$

Hint: It is helpful to recall that $\sum_{j=0}^{n-1} j = \frac{n(n-1)}{2}$ for $n \in \mathbb{N}$.

- (d) Using the above approximation, find the smallest number $n \in \mathbb{N}$ so that $\mathbf{P}[A_n] \geq \frac{1}{2}$.

- (a) Our sample space is $\Omega = \{1, 2, \dots, N\}^n$ (where $N = 365$) and $(\omega_1, \omega_2, \dots, \omega_n) \in \Omega$ stands for the outcome that person j has their birthday on the ω_j^{th} day of the year. Since we assume that every outcome of birthdays is equally likely, our probability space is then the Laplace probability space

$$(\Omega, \mathcal{P}(\Omega), \mathbf{P}), \quad \mathbf{P}[\cdot] = \frac{|\cdot|}{N^n}.$$

The event A_n^c means that no two people are born on the same day, so

$$A_n^c = \{\omega \in \Omega; \omega_i \neq \omega_j \text{ for all } 1 \leq i \neq j \leq n\}.$$

- (b) Fix $(i, j) \in \{1, \dots, n\}^2 : i \neq j$ and consider

$$B_{ij} := \{i \text{ and } j \text{ are born on the same day}\}.$$

It is not difficult to see that counting the sample points in B_{ij} corresponds exactly to counting the number of $(n-1)$ -tuplets $(\omega_1, \dots, \omega_{n-1})$ with each $\omega_k \in \{1, \dots, N\}$. Thus $|B_{ij}| = N^{n-1}$ so that

$$\mathbf{P}[B_{ij}] = \frac{N^{n-1}}{N^n} = \frac{1}{N}.$$

- (c) We have that

$$\mathbf{P}[A_n^c] = \frac{N!}{(N-n)! N^n} = \prod_{j=0}^{n-1} \left(1 - \frac{j}{N}\right).$$

Hence,

$$\log(\mathbf{P}[A_n^c]) = \sum_{j=0}^{n-1} \log\left(1 - \frac{j}{N}\right) \leq -\frac{1}{N} \sum_{j=0}^{n-1} j = -\frac{1}{N} \frac{(n-1)n}{2}.$$

This shows the upper bound

$$\mathbf{P}[A_n^c] \leq \exp\left(-\frac{n(n-1)}{2N}\right).$$

To show the lower bound, note that

$$A_n = \bigcup_{1 \leq i < j \leq n} B_{ij}.$$

where B_{ij} is the same event defined before in (b). Hence,

$$\mathbf{P}[A_n] = \mathbf{P}\left[\bigcup_{1 \leq i < j \leq n} B_{ij}\right] \leq \sum_{1 \leq i < j \leq n} \mathbf{P}[B_{ij}] = \frac{1}{N} \left(\sum_{1 \leq i < j \leq n} 1\right) = \binom{n}{2} \cdot \frac{1}{N} = \frac{n(n-1)}{2N}.$$

The latter is also equivalent to

$$1 - \frac{n(n-1)}{2N} \leq \mathbf{P}[A_n^c].$$

(d)

$$\mathbf{P}[A_n] \geq \frac{1}{2} \quad \Leftrightarrow \quad \mathbf{P}[A_n^c] \leq \frac{1}{2}.$$

Now, if $\exp\left(-\frac{n(n-1)}{2N}\right) \leq \frac{1}{2}$, then $\mathbf{P}[A_n^c] \leq \frac{1}{2}$.

$$\begin{aligned} \exp\left(-\frac{n(n-1)}{2N}\right) \leq \frac{1}{2} &\Leftrightarrow \frac{n(n-1)}{2N} \geq \log(2) \\ &\Leftrightarrow n^2 - n - 2N \log(2) \geq 0 \\ &\Leftrightarrow n \geq \frac{1 + \sqrt{1 + 8N \log(2)}}{2} =: n_1 \end{aligned}$$

With $N = 365$, we find that $n_1 \approx 22.99$ and hence $n \geq 23 \Rightarrow \mathbf{P}[A_n] \geq \frac{1}{2}$. On the other hand, if $\mathbf{P}[A_n] \geq \frac{1}{2}$ then $\mathbf{P}[A_n^c] \leq \frac{1}{2}$ and we should have

$$\begin{aligned} 1 - \frac{n(n-1)}{2N} \leq \frac{1}{2} &\Leftrightarrow \frac{n(n-1)}{2N} \geq \frac{1}{2} \\ &\Leftrightarrow n^2 - n - N \geq 0 \\ &\Leftrightarrow n \geq \frac{1 + \sqrt{1 + 4N}}{2} =: n_2 \end{aligned}$$

with $n_2 = 19.61$. Hence $\mathbf{P}[A_n] \geq \frac{1}{2} \Rightarrow n \geq 20$. Now one can explicitly compute

$$\mathbf{P}[A_{20}] \approx 0.411$$

$$\mathbf{P}[A_{21}] \approx 0.443$$

$$\mathbf{P}[A_{22}] \approx 0.475$$

$$\mathbf{P}[A_{23}] \approx 0.507$$

and find that the desired value is $n_m = 23$.

2. Bayes' theorem and the law of total probability

A poll about the use of a certain app is conducted online among three different age demographics X , Y and Z , which make up 37%, 52% and 11% of the respondents, respectively. The poll concluded that 50% of group X , 72% of group Y and 89% of group Z use the app regularly.

- Suppose we consider one randomly chosen respondent. What is the probability that the respondent uses the app regularly?
- If a respondent uses the app regularly, what is the probability that this respondent belongs to group X ?
- How many responses do we have to collect until with probability 99.9% at least one person is found that does not use the app regularly? Which suitable assumption do we have to make for this calculation?

We will use the notation:

A : person uses the app regularly,
 X : person belongs to age demographic X ,
 Y : person belongs to age demographic Y ,
 Z : person belongs to age demographic Z .

From the assignment we know $\mathbf{P}[X] = 0.37$, $\mathbf{P}[Y] = 0.52$, $\mathbf{P}[Z] = 0.11$ and $\mathbf{P}[A|X] = 0.5$, $\mathbf{P}[A|Y] = 0.72$, $\mathbf{P}[A|Z] = 0.89$.

- We use the law of total probability:

$$\begin{aligned}\mathbf{P}[A] &= \mathbf{P}[X] \cdot \mathbf{P}[A|X] + \mathbf{P}[Y] \cdot \mathbf{P}[A|Y] + \mathbf{P}[Z] \cdot \mathbf{P}[A|Z] \\ &= 0.37 \cdot 0.5 + 0.52 \cdot 0.72 + 0.11 \cdot 0.89 \\ &= 0.6573 = 65.73\%.\end{aligned}$$

- Here, we use Bayes' theorem:

$$\begin{aligned}\mathbf{P}[X|A] &= \frac{\mathbf{P}[X] \cdot \mathbf{P}[A|X]}{\mathbf{P}[X] \cdot \mathbf{P}[A|X] + \mathbf{P}[Y] \cdot \mathbf{P}[A|Y] + \mathbf{P}[Z] \cdot \mathbf{P}[A|Z]} \\ &= \frac{0.37 \cdot 0.5}{0.6573} \approx 0.28145 = 28.15\%.\end{aligned}$$

- We have to assume that the number of (possible) respondents is large enough and that the answers they give are independent. Then the probability that $n \in \mathbb{N}$ respondents use the app is given by p^n , where $p = \mathbf{P}[A]$. Therefore, we need that

$$1 - p^n \geq 0.999 \quad \Leftrightarrow \quad n \geq \frac{\log(0.001)}{\log p} = \frac{\log(0.001)}{\log(0.6573)} \approx 16.46.$$

This means that we need to collect at least 17 responses.

3. (*) Rolling dice a random number of times

[4 Points]

A fair die is rolled N times, where N is a random number with values in $\mathbb{N} = \{1, 2, \dots\}$. We call A_n the event that $N = n$ and S_M the event that the sum of all numbers shown is equal to $M \in \mathbb{N}$. We will assume the following:

- For every die roll, every number shows up with the same probability (the die is fair),
- For $n \in \mathbb{N}$, $\mathbf{P}[A_n] = \frac{1}{2^n}$.

- Determine an appropriate sample space Ω and a map

$$\mathbf{P} : \mathcal{P}(\Omega) \rightarrow [0, 1],$$

such that $(\Omega, \mathcal{P}(\Omega), \mathbf{P})$ is a probability space and (1) and (2) are fulfilled.

Calculate the probabilities

- $\mathbf{P}[A_2|S_4]$,

(c) $\mathbf{P}[S_4|A_e]$, where A_e is the event that N is an even number.

(a) We define

$$\Omega = \{(\omega_1, \dots, \omega_N) : \omega_i \in \{1, \dots, 6\}, i = 1, \dots, N, N \in \mathbb{N}\}.$$

Every element of Ω can be written as a tuple $(\omega_1, \dots, \omega_{\tilde{N}})$, for $\tilde{N} \in \mathbb{N}$, and moreover we set

$$\mathbf{P}[\{(\omega_1, \dots, \omega_{\tilde{N}})\}] = \frac{1}{6^{\tilde{N}}} \frac{1}{2^{\tilde{N}}} = \frac{1}{12^{\tilde{N}}},$$

since there are $6^{\tilde{N}}$ possibilities if the die is rolled $\tilde{N}$ times and, and the die is fair. We then put

$$\mathbf{P}[A] : \mathcal{P}(\Omega) \rightarrow [0, 1], \quad A \mapsto \sum_{a \in A} \mathbf{P}[\{a\}].$$

This is non-negative and fulfills the Kolmogorov axioms: Indeed, by our discussion in Section 3, the only part to show is the normalization. To that end, we write

$$\Omega = \bigcup_{n \in \mathbb{N}} \{(\omega_1, \dots, \omega_n) : \omega_i \in \{1, \dots, 6\}, i = 1, \dots, n\}$$

and we note that indeed

$$\begin{aligned} \mathbf{P}[\Omega] &= \sum_{n=1}^{\infty} \mathbf{P}[\{(\omega_1, \dots, \omega_n) : \omega_i \in \{1, \dots, 6\}, i = 1, \dots, n\}] \\ &= \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{1 - \frac{1}{2}} - 1 = 2 - 1 = 1, \end{aligned}$$

where we used that $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$ for any real q with $|q| < 1$.

(b) Using the definition of conditional probability (or Bayes' theorem), we find

$$\mathbf{P}[A_2|S_4] = \frac{\mathbf{P}[S_4|A_2]}{\mathbf{P}[S_4]}.$$

Note that one cannot obtain a sum of 4 when throwing more than 4 times. Therefore we see

$$S_4 = \{(4), (1, 3), (2, 2), (3, 1), (1, 1, 2), (2, 1, 1), (1, 2, 1), (1, 1, 1, 1)\}.$$

Moreover, we have

$$\begin{aligned} \mathbf{P}[S_4|A_1] &= \frac{1}{6}, \\ \mathbf{P}[S_4|A_2] &= \frac{3}{6^2}, \\ \mathbf{P}[S_4|A_3] &= \frac{3}{6^3}, \\ \mathbf{P}[S_4|A_4] &= \frac{1}{6^4}, \\ \mathbf{P}[S_4] &= \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{3}{6^2} + \frac{1}{8} \cdot \frac{3}{6^3} + \frac{1}{16} \cdot \frac{1}{6^4} \approx 0.106. \end{aligned}$$

In total, we thus find

$$\mathbf{P}[A_2|S_4] \approx \frac{\frac{3}{6^2} \cdot \frac{1}{4}}{0.106} \approx 0.197.$$

(c) We use again the definition of the conditional probability / Bayes' theorem and write

$$\mathbf{P}[S_4|A_e] = \frac{\sum_{k=1}^{\infty} \mathbf{P}[S_4|A_{2k}] \cdot \mathbf{P}[A_{2k}]}{\sum_{k=1}^{\infty} \mathbf{P}[A_{2k}]}.$$

Here, we have also used that $A_e = \bigcup_{k=1}^{\infty} A_{2k}$, and the $(A_{2k})_{k \in \mathbb{N}}$ are pairwise disjoint. Similarly as in (b), we also recognize that the numerator only consists of two summands. Therefore,

$$\mathbf{P}[S_4|A_e] = \frac{\mathbf{P}[S_4|A_2]\mathbf{P}[A_2] + \mathbf{P}[S_4|A_4]\mathbf{P}[A_4]}{\sum_{k=1}^{\infty} \frac{1}{2^{2k}}} = \frac{\frac{3}{6^2} \cdot \frac{1}{4} + \frac{1}{6^4} \cdot \frac{1}{16}}{\frac{1}{1-\frac{1}{4}} - 1} \approx 0.063.$$

4. (*) Sample size vs. signal strength

[4 Points]

An urn contains 3 red and 3 blue balls. One of these balls, called **A**, is selected uniformly at random and permanently removed from the urn without the color of this ball being shown to an observer. This observer may now draw successively – uniformly at random and with replacement – a number of individual balls (one at a time) from among the five remaining balls.

Peter may draw a ball from the urn six times, and each time the ball he draws turns out to be red. Paula may draw a ball from the urn 600 times; 303 times she draws turns out to be red, and 297 times she draws a blue ball.

Let R be the event that **A** is red, P_1 be the event that Peter's sequence is realized, and P_2 be Paula's sequence is realized.

- (a) Calculate $\mathbf{P}[R]$, $\mathbf{P}[R|P_1]$, and $\mathbf{P}[R|P_2]$.
- (b) Interpret your result.

(a) We have

$$\mathbf{P}[R] = \frac{3}{3+3} = \frac{1}{2}.$$

By Bayes' theorem, for $i = 1, 2$,

$$\mathbf{P}[R|P_i] = \frac{\mathbf{P}[P_i|R] \cdot \mathbf{P}[R]}{\mathbf{P}[P_i|R] \cdot \mathbf{P}[R] + \mathbf{P}[P_i|R^c] \cdot \mathbf{P}[R^c]} = \frac{1}{1 + \frac{\mathbf{P}[P_i|R^c]}{\mathbf{P}[P_i|R]}},$$

where we used $\mathbf{P}[R] = \mathbf{P}[R^c]$ in the last equality. As

$$\mathbf{P}[P_1|R] = \frac{2^6}{5^6} = \left(\frac{2}{5}\right)^6, \quad \mathbf{P}[P_1|R^c] = \left(\frac{3}{5}\right)^6,$$

we obtain $\mathbf{P}[R|P_1] = \frac{1}{1+(3/2)^6} \approx 0.081$. Similarly, as

$$\mathbf{P}[P_2|R] = \frac{\binom{600}{303} 2^{303} 3^{297}}{5^{600}} = \binom{600}{303} \left(\frac{2}{5}\right)^{303} \left(\frac{3}{5}\right)^{297}, \quad \mathbf{P}[P_2|R^c] = \binom{600}{303} \left(\frac{3}{5}\right)^{303} \left(\frac{2}{5}\right)^{297},$$

we obtain $\mathbf{P}[R|P_2] = \frac{1}{1+(3/2)^6} \approx 0.081$.

- (b) The same will be true whenever the difference between red and blue balls drawn is the same for Peter and Paula. Given that difference – the “signal strength” – the number of draws in which it was obtained does not differentially influence the strength of the evidence obtained.